introduction

Cybersecurity is an important part of protecting computer systems, networks, applications, and organizational information from security incidents and operational problems.

This project focuses on a cybersecurity incident related to the deployment of a critical software patch.

The selected scenario is **“The Quick Fix.”** In this scenario, a critical patch was deployed without testing. After the deployment, users reported that they could no longer log in to the organization’s systems.

This project analyzes the incident, identifies the weakness in the patch and change management process, and proposes preventive measures and security policies.

Project Scope

This project focuses on analyzing a cybersecurity incident related to the deployment of a critical patch. It also presents a proposed network topology using Cisco Packet Tracer and performs baseline connectivity testing. The project discusses the identified vulnerability, preventive measures, security policies, recommendations, and an estimated budget for implementing the proposed security measures.

Why Cyber Security Matters?

Cybersecurity is important because organizations depend on computer systems and networks to provide services and manage important information.

A problem in one system can affect users and services. For this reason, organizations need effective processes to identify problems, respond to incidents, and reduce the risk of similar incidents.

In the selected scenario, a critical patch was deployed without testing, and users later reported that they could no longer log in to the organization’s systems.

This demonstrates the importance of testing and controlling major changes before applying them to production systems.

Importance to the Organization

Cybersecurity is important because security incidents can affect the availability and reliability of organizational systems. A poorly controlled change can interrupt normal services and prevent users from accessing important systems. Applying proper security processes, testing, monitoring, and change management can help reduce these risks and support the continuity of organizational operations.

Scenario Description

The selected scenario is **“The Quick Fix.”**

In this scenario, Joe, a network administrator, is asked to deploy a critical patch. He quickly builds an installation file and deploys it before leaving.

After the deployment, Sue receives calls reporting that nobody can log in to the organization’s systems. The scenario states that no testing was performed for the recently installed critical patch.

The main incident investigated in this project is the users’ inability to log in after the patch deployment. The patch should be investigated as a possible cause of the problem.

Scenario Analysis

The scenario demonstrates how an urgent technical change can affect normal system operations when it is introduced without sufficient testing. In this case, the critical patch was deployed before adequate testing, and users subsequently reported login problems. The scenario highlights the importance of controlled changes, testing, documentation, and verification before deploying important updates.

Proposed Network Topology

The original scenario does not specify a network topology. Therefore, a proposed network topology was designed for this project using Cisco Packet Tracer to represent a simple organizational network.

The proposed topology consists of a Cloud, Router, Switch, PC, and Server.

The PC represents a user workstation, while the Server represents an organizational server. The Switch connects the internal devices, and the Router provides the network gateway.

The proposed topology is designed to demonstrate the network environment and perform baseline connectivity tests before applying the simulated change.

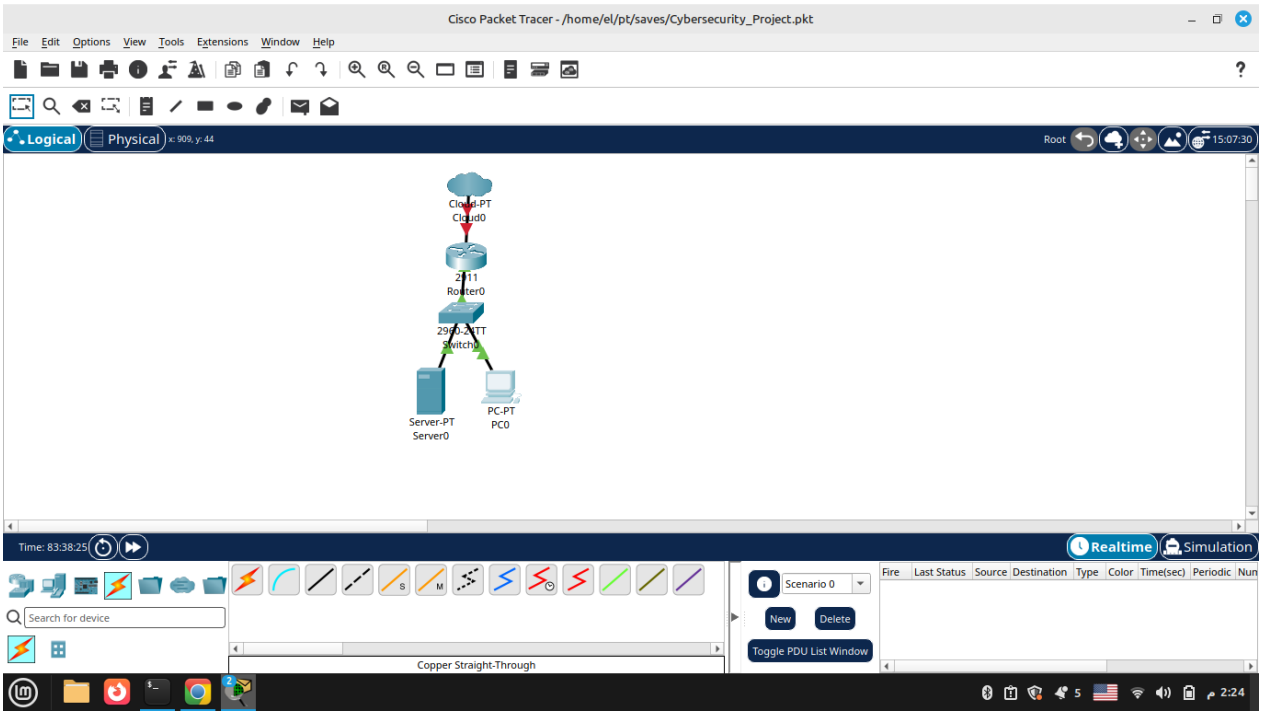


Figure 1: Proposed network topology implemented using Cisco Packet Tracer.

Components and Description

The proposed network contains several components that work together to provide basic network connectivity.

Component	Description
Cloud	Represents the external network environment in the Packet Tracer simulation.
Router	Provides network routing and acts as the gateway for the internal network.
Switch	Connects the PC, Server, and Router within the internal network.
PC	Represents a user's workstation in the organization.
Server	Represents an organizational server used in the proposed network.
Network Cables	Provide the physical connections between the simulated devices.

Practical Role of the Components

The components in the proposed topology work together to provide basic network connectivity. The router provides the gateway for the internal network, while the switch connects the internal devices. The PC represents a user workstation, and the server represents an organizational server. The cloud represents the external network environment used in the proposed topology.

IP Addressing

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.1.1	255.255.255.0	—
PC	192.168.1.10	255.255.255.0	192.168.1.1
Server	192.168.1.20	255.255.255.0	192.168.1.1

IP Addressing Explanation

The IP addressing scheme was designed to place the router, PC, and server in the same internal network. The router uses **192.168.1.1** as the gateway, while the PC uses **192.168.1.10** and the Server uses **192.168.1.20**. The subnet mask **255.255.255.0** is used for the internal network.

Practical Implementation – Baseline Testing

Baseline Test:

Before applying the critical patch, connectivity between the PC, Server, and Router was verified successfully. Ping tests showed 0% packet loss.

Test	Source	Destination	Result
Test 1	PC	Router 192.168.1.1	Successful – 0% packet loss
Test 2	PC	Server 192.168.1.20	Successful – 0% packet loss
Test 3	Server	Router 192.168.1.1	Successful – 0% packet loss

Incident Description

A critical patch was deployed without testing. After the deployment, users reported that they could no longer log in to the organization's systems.

Response Plan

The incident should be handled in a controlled manner .First ,the affected systems should be identified and the incident should be documented .The deployment of the patch should be reviewed ,and the organization should determine whether the patch is related to the login problem .Testing should be performed before any further deployment.

Incident Response Process

The incident response process should be performed in a controlled and organized manner. The organization should first identify the affected systems and document the reported login problem.

The recently deployed critical patch should then be reviewed to determine whether it may be related to the incident. The organization should perform appropriate testing before making additional changes.

After identifying the cause or contributing factors, the organization should apply the appropriate corrective action and verify that the affected services are working correctly.

Finally, the incident and the actions taken should be documented for future review and to help prevent similar incidents.

Practical Testing

The proposed network was implemented using Cisco Packet Tracer to verify basic connectivity between the network devices.

The PC was configured with the IP address **192.168.1.10**, while the Server was configured with **192.168.1.20**. The Router interface connected to the internal network was configured with **192.168.1.1**.

Connectivity tests were performed using ping commands. The PC successfully communicated with the Router and Server, and the Server successfully communicated with the Router.

The tests showed **0% packet loss**, confirming that the proposed internal network was functioning correctly during the baseline test.

Test Results

Test	Source	Destination	Result
Ping Test 1	PC	Router (192.168.1.1)	Successful – 0% packet loss
Ping Test 2	PC	Server (192.168.1.20)	Successful – 0% packet loss
Ping Test 3	Server	Router (192.168.1.1)	Successful – 0% packet loss

The test results show that the main devices in the proposed internal network were able to communicate successfully. These results provide a baseline for comparing the network before and after any future changes

Network Security Controls

Network security controls can help protect the proposed organizational network from unauthorized access and reduce the impact of security incidents. The controls should be applied according to the organization's requirements and the role of each network component.

Access to network devices should be restricted to authorized users. Administrative credentials should be protected, and unnecessary services should be disabled when they are not required.

The organization should also monitor important systems and network activity. Monitoring can help identify unusual behavior and provide useful information during incident investigation.

Regular backups and documented recovery procedures should be maintained for important systems. In addition, critical changes should follow the approved change-management process and should be tested before deployment.

These controls complement the patch management and change management procedures discussed in this project and provide additional protection for the proposed network.

Network Monitoring

Network monitoring is an important part of maintaining the security and reliability of an organizational network. Monitoring can help identify unusual activity, connectivity problems, and unexpected changes that may affect users or services.

In the proposed network, the organization should monitor the router, switch, server, and other important network devices. Network administrators should review relevant logs and investigate unexpected events.

Monitoring should also be performed after deploying critical patches or other important changes. This can help the organization identify problems quickly and take corrective action before the impact becomes larger.

The monitoring process should be documented so that information collected during an incident can be reviewed later and used to improve future security procedures.

Lessons Learned

This project demonstrates the importance of testing and controlling critical system changes before deployment. The incident showed that a change introduced without adequate testing can result in problems for users and organizational services.

One important lesson is that critical patches should follow a documented process that includes testing, approval, deployment, verification, and rollback planning.

Another lesson is the importance of baseline testing. Establishing normal network connectivity before making changes provides a reference that can be used when investigating problems after a change.

Finally, documenting incidents and the actions taken can help the organization improve its procedures and reduce the possibility of similar problems occurring in the future.

The following checklist can be used before deploying a critical patch or making another important system change:

1. Identify the systems and services that may be affected.
2. Review the purpose and importance of the planned change.
3. Test the patch or change in a controlled environment.
4. Document the test results and any problems identified.
5. Obtain the required approval before production deployment.
6. Prepare a rollback plan before applying the change.
7. Deploy the change in a controlled manner.
8. Verify important services after deployment.
9. Monitor the systems for unexpected problems.
10. Document the final results and any lessons learned.

This checklist provides a practical way to apply the patch management and change management principles discussed in this project.

References

1. Project Scenario: “The Quick Fix” – Amait Learning Cybersecurity Project.
2. Cisco Packet Tracer – Used for network topology design and connectivity testing.
3. Project Notes and Practical Testing Results – Cisco Packet Tracer baseline tests.

Appendix A – Packet Tracer Evidence

This appendix provides practical evidence of the network implementation performed using Cisco Packet Tracer. The topology includes the Router, Switch, PC, and Server used in the project.

The configured IP addresses were used to establish communication between the network devices. Connectivity was verified using ping tests, and the results showed successful communication with 0% packet loss.

The screenshots included in this appendix document the practical implementation and testing performed during the project.

Appendix B – Test Evidence

The connectivity tests performed during the practical implementation were used to verify the baseline condition of the proposed network.

The PC successfully reached the Router at **192.168.1.1** with **0% packet loss**. The PC also successfully reached the Server at **192.168.1.20** with **0% packet loss**.

These results confirm that the configured devices were able to communicate successfully during the baseline testing stage. The results can be used as a reference when evaluating the network after future changes.

Test Summary:

- Router 192.168.1.1 — Successful — 0% packet loss.
- Server 192.168.1.20 — Successful — 0% packet loss.

Appendix C – Project Summary

Project Area	What Was Completed
Scenario Analysis	Analyzed the critical patch deployment incident and login problems.
Network Topology	Designed and implemented the proposed topology in Cisco Packet Tracer.
IP Addressing	Configured the Router, PC, and Server using the 192.168.1.0/24 network.
Baseline Testing	Performed connectivity tests using Ping.
Vulnerability	Identified weaknesses in patch management and change management.
Prevention	Proposed testing, verification, documentation, and rollback procedures.

Security Policies	Covered Patch Management, Change Management, Testing and Verification, and Rollback.
Recommendation	Recommended controlled deployment and continuous review of the change process.
Budget	Provided estimated costs for the proposed security measures.
Lessons Learned	Highlighted the importance of testing and controlling critical changes.

Risk Analysis

The selected scenario demonstrates that a change to an important system can create operational problems when it is introduced without adequate testing.

The main risk identified in this project is related to the deployment of a critical patch without prior testing. The resulting login problem can affect users' ability to access organizational systems.

The risk can be reduced by testing patches in a controlled environment, following change-management procedures, verifying systems after deployment, and preparing a rollback plan before implementing critical changes.

These measures help the organization reduce the impact of unexpected problems and improve the control of future system changes.

Conclusion

This project analyzed the “The Quick Fix” cybersecurity scenario, in which a critical patch was deployed without testing and users subsequently reported login problems.

The project identified the main weakness as a failure in the patch management and change management process. A proposed network topology was also implemented in Cisco Packet Tracer, and baseline connectivity tests were successfully completed with 0% packet loss.

The proposed security measures focus on controlled testing, proper change management, documentation, verification, monitoring, and rollback planning.

These measures provide a structured approach for reducing the risk of similar incidents and improving the management of critical system changes.

Vulnerability Categorization

The main vulnerability in this scenario is the failure to test the critical patch before deployment. The patch was deployed quickly without adequate testing, which was followed by user login problems. This represents a weakness in the patch management and change management process

The vulnerability in this scenario can be categorized as a weakness in the patch management and change management process. The main issue is not simply the existence of a critical patch, but the way the patch was handled before deployment.

The patch was deployed without adequate testing, and users subsequently reported that they could no longer log in to the organization's systems. This shows how an uncontrolled change can create an operational impact on users and services.

The incident also shows the importance of having a controlled testing process before applying important changes to production systems. Testing can help identify unexpected problems before the change affects normal users.

Therefore, the project focuses on improving the process used to test, approve, deploy, and verify critical patches rather than treating the incident as only a technical problem.

How to Prevent Vulnerabilities

To prevent similar incidents, the organization should test critical patches in a controlled environment before deploying them to production systems. Patch deployment should follow an approved change-management process, with proper documentation, testing, and verification before and after deployment.

Security Policies

1. Patch Management Policy:
All critical patches must be tested in a controlled environment before production deployment.

2. Change Management Policy:
Changes to production systems must be documented, reviewed, and approved before implementation.

3. Testing and Verification Policy:
Systems must be tested before and after major changes to verify that essential services remain available.

4. Rollback Policy:
A rollback plan should be prepared before deploying a critical patch, so the organization can return to the previous configuration if problems occur.

These security policies should be applied consistently to reduce the possibility of similar incidents. Patch management should define how critical patches are tested, approved,

deployed, and verified. Change management should ensure that important changes are reviewed and documented before they are introduced into production.

Testing and verification should be performed before and after significant changes. In addition, a rollback procedure should be available so that the organization can respond if a change produces unexpected results.

Following these policies provides a structured process for managing critical changes and helps the organization maintain more reliable and controlled systems.

Recommendation

The organization should improve its patch and change management processes. Critical patches should be tested in a controlled environment before deployment to production systems. The organization should also maintain proper documentation, approval procedures, verification steps, and a rollback plan for critical changes. Regular monitoring after deployment should be performed to identify any unexpected problems quickly.

The organization should also review the effectiveness of its patch and change management procedures regularly. Lessons learned from incidents should be documented and used to improve future deployment processes.

Critical changes should have clearly defined responsibilities, approval steps, testing requirements, and verification procedures. This can help ensure that important system changes are introduced in a controlled and traceable manner.

Budget

Item	Purpose	Estimated Cost
Patch Testing Environment	Testing critical patches before deployment	$500
Backup Solution	Backup and recovery before major changes	$300
Monitoring Tools	Monitoring systems after deployment	$400
Staff Training	Training employees on patch and change management	$300
Documentation and Procedures	Creating and maintaining security procedures	$100
Total		**$1,600**

Note: The costs shown in this budget are estimated values used for the project to demonstrate the resources that may be required to implement the proposed security measures. They are not actual market quotations.